

Lab-6 (Structure and Union)

1. Create a structure named "student" with members name, age, and total marks and wap to input data of two input data of two students and display their information.

Ans:

```
#include<stdio.h>
#include<string.h>
struct student{
    char name[50];
    int age;
    float mark;
};
int main()
{
    struct student a,b;
    printf("Enter name of first student:");
    fgets(a.name,50,stdin);
    a.name[strlen(a.name)-1]='\0';
    printf("Enter age:");
    scanf("%d",&a.age);
    printf("Enter total marks:");
    scanf("%f",&a.mark);

    printf("Enter name of first student:");
    scanf(" %50[^\n]s",b.name);
    printf("Enter age:");
    scanf("%d",&b.age);
    printf("Enter total marks:");
    scanf("%f",&b.mark);

    printf("The record of two student are:\n");
    printf("%s\t%d\t%f\t",a.name,a.age,a.mark);
    printf("\n");
    printf("%s\t%d\t%f\t",b.name,b.age,b.mark);
    return 0;
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\structure and union.exe

```
Enter name of first student:Aayush Baral
Enter age:20
Enter total marks:450
Enter name of first student:Baral Aayush
Enter age:19
Enter total marks:400
The record of two student are:
Aayush Baral    20    450.000000
Baral Aayush    19    400.000000
-----
Process exited after 24.21 seconds with return value 0
Press any key to continue . . .
```

2. Defined structure named circle with member radius and wap to calc area and perimeter of two circle and display result.

Ans:

```
#include<stdio.h>
```

```
struct circle{
    float radius;
}c[2];
```

```
int main()
```

```
{
    int i;
    float area,per;
    for(i=0; i<2; i++)
    {
        printf("Enter radius of %dst circle:",i);
        scanf("%f",&c[i].radius);
    }

    for(i=0; i<2; i++)
    {
        area=3.1415*c[i].radius*c[i].radius;
        per=2*3.1415*c[i].radius;
        printf("The area and perimeter of %dst circle is:",i);
        printf("%f,%f",area,per);
        printf("\n");
    }
    return 0;
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\structure and union.exe

```
Enter radius of 0st circle:2
Enter radius of 1st circle:4
The area and perimeter of 0st circle is:12.566000,12.566000
The area and perimeter of 1st circle is:50.264000,25.132000

-----
Process exited after 8.78 seconds with return value 0
Press any key to continue . . .
```

3. Write using structure that reads record of 5 student with members roll, name, address and marks. And display the record of student who have obtained greater than 250 marks.

Ans:

```
#include<stdio.h>
#include<string.h>
```

```
struct student{
    char name[50];
    char address[50];
    int roll;
    float mark;
}s[5];

int main()
{
    int i;
    for(i=0; i<5; i++)
    {
        printf("Enter name of student:");
        scanf(" %50[^\n]s",s[i].name);

        printf("Enter address:");
        scanf(" %50[^\n]s",s[i].address);

        printf("Enter roll number:");
        scanf("%d",&s[i].roll);

        printf("Enter mark:");
        scanf("%f",&s[i].mark);

        printf("\n");
    }

    printf("The output of students are:\n");
```

```

printf("Name\t Address\t Roll\t Mark\n");
for(i=0; i<5; i++)
{
    if(s[i].mark>250)
printf("%s\t%s\t%d\t%f\n",s[i].name,s[i].address,s[i].roll,s[i].mark);
}
return 0;
}

```

Output:  C:\Users\ayus\OneDrive\Desktop\New folder\C-program\structure and union.exe

```

Enter name of student:shyam
Enter address:ktm
Enter roll number:02
Enter mark:270

Enter name of student:sita
Enter address:pok
Enter roll number:03
Enter mark:300

Enter name of student:rita
Enter address:bir
Enter roll number:04
Enter mark:320

Enter name of student:gita
Enter address:jan
Enter roll number:05
Enter mark:200

The output of students are:
Name      Address      Roll      Mark
shyam     ktm          2         270.000000
sita      pok          3         300.000000
rita      bir          4         320.000000

-----
Process exited after 101.7 seconds with return value 0
Press any key to continue . . .

```

4. Create a structure named book with members book_name, price and author-name then take input for 10 records of book and then print the name of author having the price of book greater than 1000.

Ans:

```

#include<stdio.h>
#include<string.h>
struct book{
    char book_name[50];

```

```

        char author_name[50];
        int price;
    }b[2];
int main()
{
    int i;
    for(i=0; i<2; i++)
    {
        printf("Enter name of book:");
        scanf(" %50[^\n]s",b[i].book_name);
        printf("Enter author name:");
        scanf(" %50[^\n]s",b[i].author_name);
        printf("Enter price:");
        scanf("%d",&b[i].price);
        printf("\n");

    }

    printf("The author of book are:\n");
    for(i=0; i<2; i++)
    {
        if(b[i].price>1000)
        {
            printf("%s",b[i].author_name);

        }
    }
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\structure and union.exe

```

Enter name of book:Mathematics
Enter author name:Lekhnath paudel
Enter price:1100

Enter name of book:Science
Enter author name:Chitra kc
Enter price:1000

Enter name of book:Social
Enter author name:Santosh Baral
Enter price:1200

Enter name of book:English
Enter author name:Binod Jha
Enter price:1020

Enter name of book:Nepali
Enter author name:Chabilal Nepali
Enter price:900

The author of book are:
Lekhnath paudelSantosh BaralBinod Jha
-----
Process exited after 144.4 seconds with return value 0
Press any key to continue . . .

```

5. Create a union named person with members name,age and salary. Then input data of 2 person and print output.

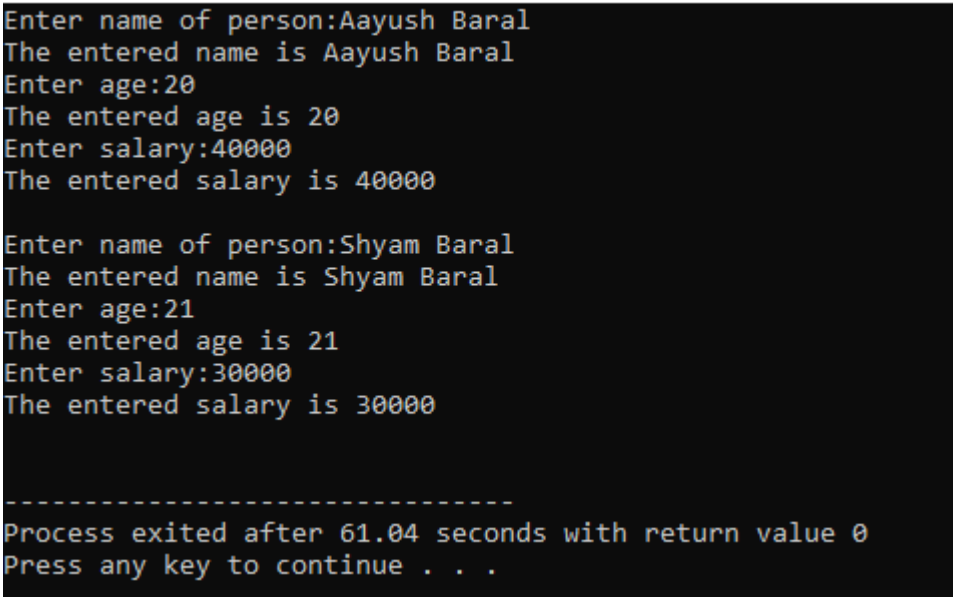
Ans:

```
#include<stdio.h>
#include<string.h>
union person{
    char name[50];
    int age;
    int salary;
}p[2];
int main()
{
    int i;
    for(i=0; i<2; i++)
    {
        printf("Enter name of person:");
        scanf(" %50[^\n]s",p[i].name);
        printf("The entered name is %s\n",p[i].name);

        printf("Enter age:");
        scanf("%d",&p[i].age);
        printf("The entered age is %d\n",p[i].age);

        printf("Enter salary:");
        scanf("%d",&p[i].salary);
        printf("The entered salary is %d\n",p[i].salary);
        printf("\n");
    }
    return 0; }
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\structure and union.exe



```
Enter name of person:Aayush Baral
The entered name is Aayush Baral
Enter age:20
The entered age is 20
Enter salary:40000
The entered salary is 40000

Enter name of person:Shyam Baral
The entered name is Shyam Baral
Enter age:21
The entered age is 21
Enter salary:30000
The entered salary is 30000

-----
Process exited after 61.04 seconds with return value 0
Press any key to continue . . .
```

6. Create a structure named time with members hrs, min, sec. input two times and display the sum.

Ans:

```
#include<stdio.h>
struct time{
    int hr;
    int min;
    int sec;
}t[2];

int main()
{
    int i,sum=0;
    for(i=0; i<2; i++)
    {
        printf("Enter hour:");
        scanf("%d",&t[i].hr);

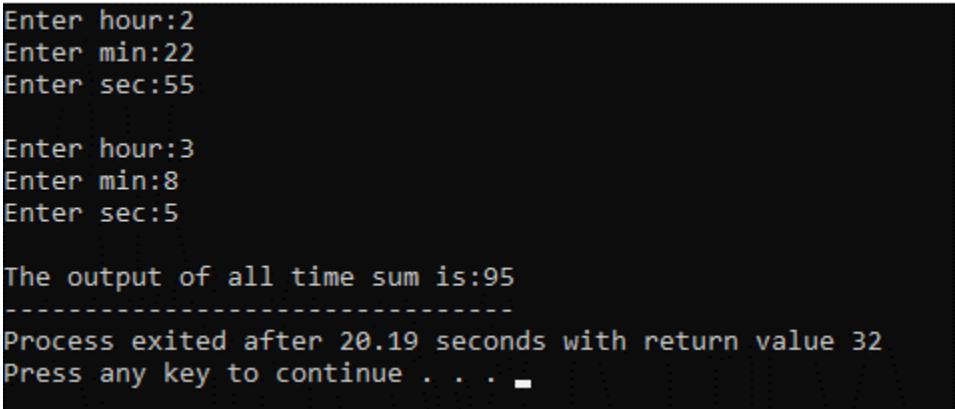
        printf("Enter min:");
        scanf("%d",&t[i].min);

        printf("Enter sec:");
        scanf("%d",&t[i].sec);

        sum=sum+t[i].hr+t[i].min+t[i].sec;

        printf("\n");
    }
    printf("The output of all time sum is:%d",sum);
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\structure and union.exe



```
Enter hour:2
Enter min:22
Enter sec:55

Enter hour:3
Enter min:8
Enter sec:5

The output of all time sum is:95
-----
Process exited after 20.19 seconds with return value 32
Press any key to continue . . .
```

7. Write a c-program to find sum of two complex number input by user using structure.

Ans:

```
#include<stdio.h>
struct complex{
    int real;
    int img;
}a,b;

int main()
{
    int sum_real,sum_img;
    printf("Enter first complex number real part:");
    scanf("%d",&a.real);

    printf("Enter first complex number imaginary part:");
    scanf("%d",&a.img);

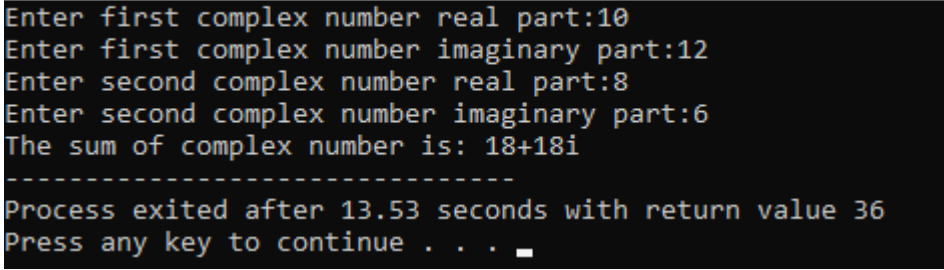
    printf("Enter second complex number real part:");
    scanf("%d",&b.real);

    printf("Enter second complex number imaginary part:");
    scanf("%d",&b.img);

    sum_real=a.real+b.real;
    sum_img=a.img+b.img;

    printf("The sum of complex number is: %d+%di",sum_real,sum_img);
}
```

Output:  C:\Users\ayus\OneDrive\Desktop\New folder\C-program\structure and union.exe



```
Enter first complex number real part:10
Enter first complex number imaginary part:12
Enter second complex number real part:8
Enter second complex number imaginary part:6
The sum of complex number is: 18+18i
-----
Process exited after 13.53 seconds with return value 36
Press any key to continue . . .
```